

2018-12-18 TRAFFIC ANALYSIS EXERCISE - ANSWERS

<https://www.malware-traffic-analysis.net/2018/12/18/index.html>

LAN segment details:

- LAN segment range: 172.16.3.0/24 (172.16.3.0 through 172.16.3.255)
- Domain: eggnogsoup.com
- Domain Controller (DC): 172.16.3.2 - EggNogSoup-DC
- LAN segment gateway: 172.16.3.1
- LAN segment broadcast address: 172.16.3.255

Tasks:

- How many hosts besides the DC at 172.16.3.2 are active on the network?
- List the IP addresses for the hosts found when investigating the previous question.
- Which IP address represents a host running Ubuntu?
- What type of host is using IP address 172.16.3.188?
- Which IP address is mostly likely an Amazon Fire tablet?
- Which three IP addresses represent Windows hosts that connect to the domain controller at 172.16.3.2?
- Which of the three Windows hosts shows indications of an infection with Emotet and IcedID banking Trojan (Bokbot)?
- Which IP address is a host running Android 8.0.0?
- What is the brand and model of the phone running Android 8.0.0?
- What is the brand and type of device on 172.16.3.112?

ANSWERS:

Q: How many hosts besides the DC at 172.16.3.2 are active on the network?

A: **9**

Q: List the IP addresses for the hosts found when investigating the previous question.

A: **172.16.3.109**

172.16.3.110

172.16.3.111

172.16.3.112

172.16.3.114

172.16.3.122

172.16.3.133

172.16.3.188

172.16.3.189

Q: Which IP address represents a host running Ubuntu?

A: **172.16.3.110**

Q: What type of host is using IP address 172.16.3.188?

A: **Apple iPhone**

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Q: Which IP address is mostly likely an Amazon Fire tablet?

A: **172.16.3.109**

Q: Which three IP addresses represent Windows hosts that connect to the domain controller at 172.16.3.2?

A: **172.16.3.114, 172.16.3.133, and 172.16.3.189**

Q: Which of the three Windows hosts shows indications of an infection with Emotet and IcedID banking Trojan (Bokbot)?

A: **172.16.3.133**

Q: Which IP address is a host running Android 8.0.0?

A: **172.16.3.111**

Q: What is the brand and model of the phone running Android 8.0.0?

A: **Moto E5 Play**

Q: What is the brand and type of device on 172.16.3.112?

A: **Samsung Galaxy Tab E Lite**

DETAILS:

As always before doing these exercises, I recommend you customize your column display in Wireshark. I've written a guide here:

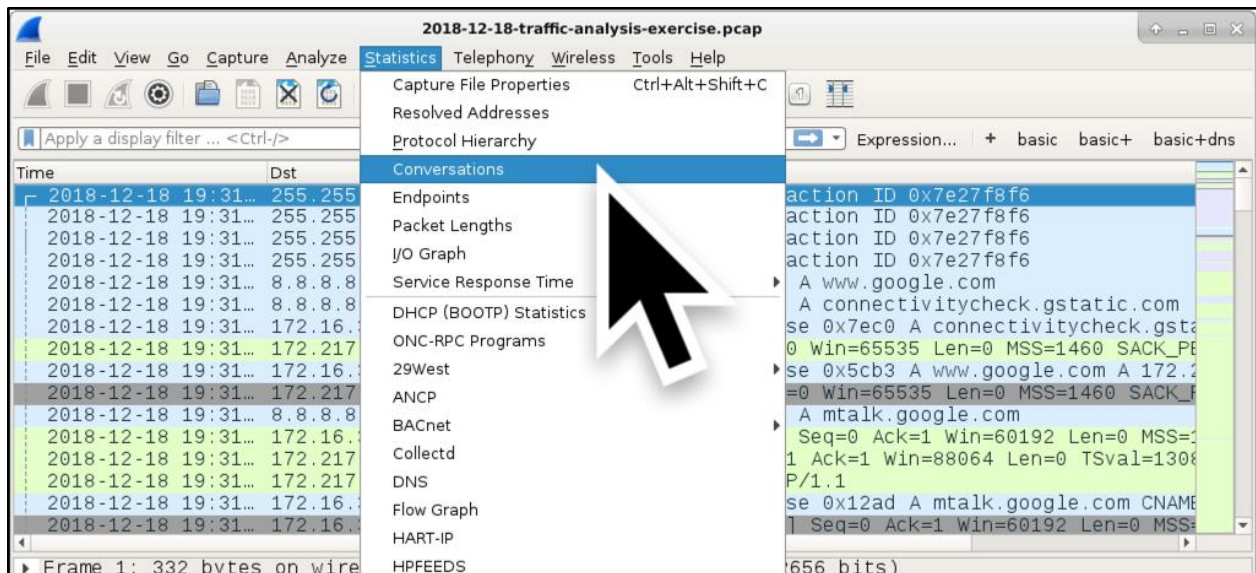
- <https://researchcenter.paloaltonetworks.com/2018/08/unit42-customizing-wireshark-changing-column-display/>

Q: How many hosts besides the DC at 172.16.3.2 are active on the network?

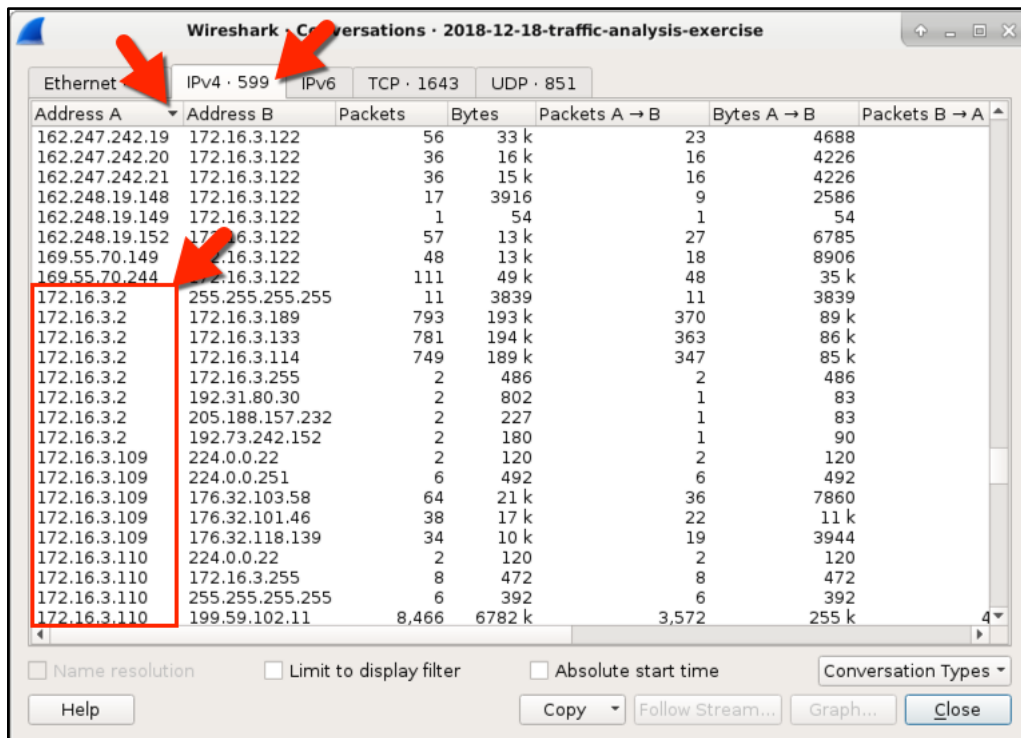
Q: List the IP addresses for the hosts found when investigating the previous question.

Details: Since we know from the LAN segment data that any hosts (clients) on the internal network will be on 172.16.3.0/24, we can quickly use the Wireshark menu path for **Statistics → Conversations** to find the answers. Once you bring up the Conversations menu, go to the IPv4 tab and sort on Address A. Then scroll down to where you start seeing IP addresses starting with **172.16.3.** and document each of the different IP addresses listed.

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Shown above: Getting to the Conversations window in Wireshark.

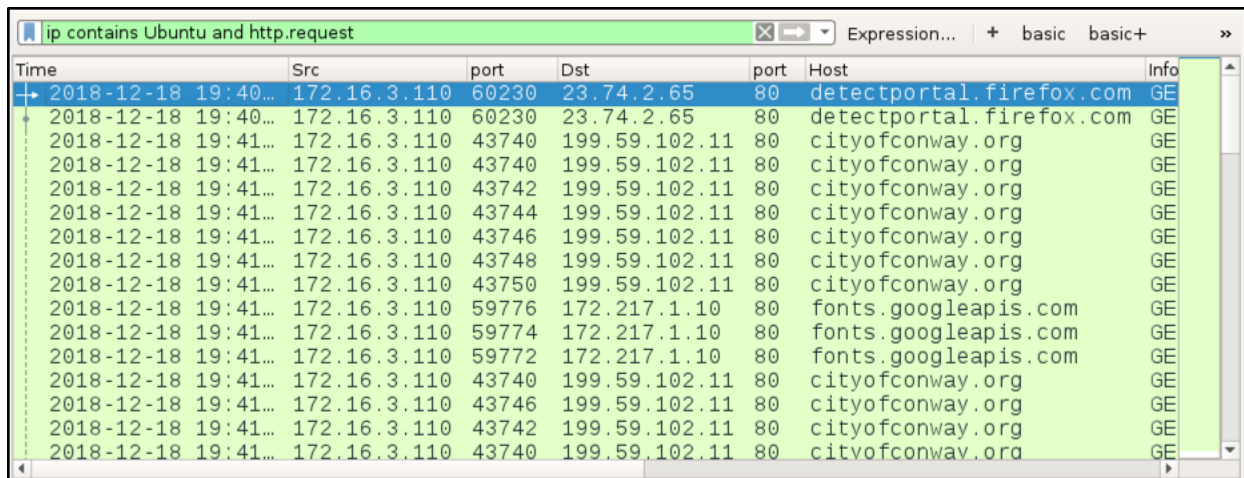


Shown above: Finding IPv4 addresses in the 172.16.3.0/24 range.

Q: Which IP address represents a host running Ubuntu?

Details: A quick way to check this is to search for HTTP traffic with "Ubuntu" listed in the HTTP request headers. Use **ip contains Ubuntu** and **http.request** (this search is case-sensitive, so be sure to capitalize "Ubuntu"). The results show HTTP requests generated by **172.16.3.110**. Follow TCP stream for any of the first few frames listed in those search results. You'll see Ubuntu listed in the User-Agent string in the HTTP request headers. Ubuntu is quite proud to identify itself in HTTP traffic.

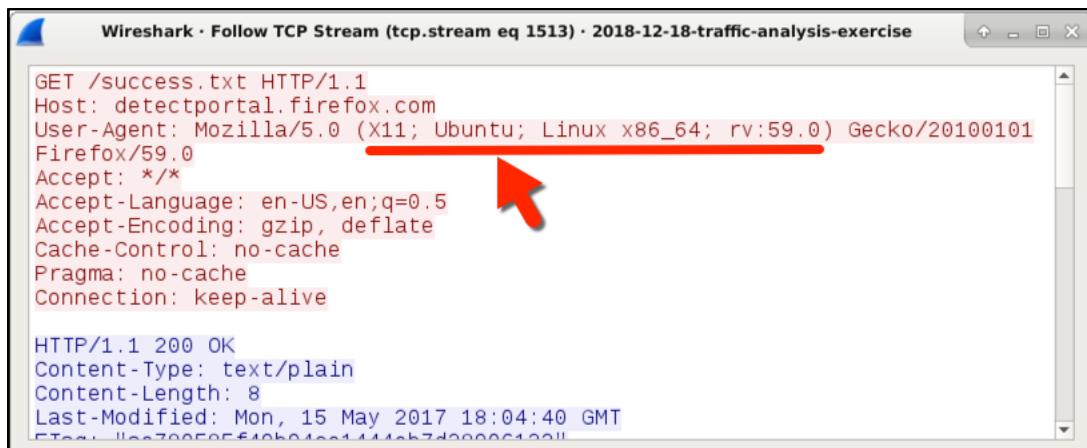
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The screenshot shows the Wireshark packet list with a filter expression 'ip contains Ubuntu and http.request'. The table displays 15 filtered packets, all originating from 172.16.3.110 and destined for various hosts including detectportal.firefox.com, cityofconway.org, and fonts.googleapis.com. The 'Info' column for each packet shows 'GET' requests.

Time	Src	port	Dst	port	Host	Info
2018-12-18 19:40...	172.16.3.110	60230	23.74.2.65	80	detectportal.firefox.com	GE
2018-12-18 19:40...	172.16.3.110	60230	23.74.2.65	80	detectportal.firefox.com	GE
2018-12-18 19:41...	172.16.3.110	43740	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43740	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43742	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43744	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43746	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43748	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43750	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	59776	172.217.1.10	80	fonts.googleapis.com	GE
2018-12-18 19:41...	172.16.3.110	59774	172.217.1.10	80	fonts.googleapis.com	GE
2018-12-18 19:41...	172.16.3.110	59772	172.217.1.10	80	fonts.googleapis.com	GE
2018-12-18 19:41...	172.16.3.110	43740	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43746	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43742	199.59.102.11	80	cityofconway.org	GE
2018-12-18 19:41...	172.16.3.110	43740	199.59.102.11	80	cityofconway.org	GE

Shown above: Filtering for HTTP requests that include "Ubuntu" as a string of characters.



The screenshot shows the 'Follow TCP Stream' window for the selected packet. It displays the raw HTTP request and response. A red arrow points to the 'User-Agent' line, which identifies the client as 'Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:59.0) Gecko/20100101 Firefox/59.0'.

```
GET /success.txt HTTP/1.1
Host: detectportal.firefox.com
User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:59.0) Gecko/20100101 Firefox/59.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Cache-Control: no-cache
Pragma: no-cache
Connection: keep-alive

HTTP/1.1 200 OK
Content-Type: text/plain
Content-Length: 8
Last-Modified: Mon, 15 May 2017 18:04:40 GMT
ETag: "f40b04ee144eb7d00000000000000000"
```

Shown above: Confirming the HTTP traffic is from an Ubuntu host.

Q: What type of host is using IP address 172.16.3.188?

Details: If you filter on `ip.addr eq 172.16.3.188`, you'll find DNS requests to Apple-related domains, and you'll see the MAC address is an Apple MAC address. If there's regular HTTP traffic, we can easily find out what type of Apple device this is, so filter on `ip.addr eq 172.16.3.188 and http.request`. Follow the HTTP request to `nymag.com`. In it, you'll see this is an iPhone based on the User-Agent line from the HTTP GET request headers.

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ip.addr eq 172.16.3.188

	Src	port	Dst	port	Info
19:34...	172.16.3.188	59887	8.8.8.8	53	Standard query 0x03ee A setup.icloud.com
19:34...	8.8.8.8	53	172.16.3.188	59...	Standard query response 0x03ee A setup.i
19:34...	172.16.3.188	53503	17.248.138.111	443	53503 → 443 [SYN, ECN, CWR] Seq=0 Win=65535
19:34...	17.248.138.111	443	172.16.3.188	53...	443 → 53503 [SYN, ACK, ECN] Seq=0 Ack=1
19:34...	172.16.3.188	63387	8.8.8.8	53	Standard query 0x46b7 A www.apple.com
19:34...	172.16.3.188	64158	8.8.8.8	53	Standard query 0x91a0 A www.icloud.com
19:34...	172.16.3.188	53503	17.248.138.111	443	53503 → 443 [ACK] Seq=1 ACK=1 Win=131328
19:34...	172.16.3.188	53503	17.248.138.111	443	Client Hello
19:34...	172.16.3.188	58262	8.8.8.8	53	Standard query 0x9f44 A time-ios.apple.c
19:34...	17.248.138.111	443	172.16.3.188	53...	443 → 53503 [ACK] Seq=1 ACK=248 Win=3004
19:34...	17.248.138.111	443	172.16.3.188	53...	Server Hello
19:34...	17.248.138.111	443	172.16.3.188	53...	[TCP segment of a reassembled PDU]
19:34...	8.8.8.8	53	172.16.3.188	64...	Standard query response 0x91a0 A www.icl
19:34...	8.8.8.8	53	172.16.3.188	63...	Standard query response 0x46b7 A www.app
19:34...	172.16.3.188	59153	8.8.8.8	53	Standard query 0xaa66 A init-p01st.push.
19:34...	172.16.3.188	53503	17.248.138.111	443	53503 → 443 [ACK] Seq=248 Ack=2553 Win=1

Frame 29471: 111 bytes on wire (888 bits), 111 bytes captured (888 bits)

Ethernet II, Src: Cisco_a2:bf:89 (00:03:31:a2:bf:89), Dst: Apple_c1:99:a5 (00:26:08:c1:99:a5)

Internet Protocol Version 4, Src: 8.8.8.8, Dst: 172.16.3.188

User Datagram Protocol, Src Port: 53, Dst Port: 59153

Domain Name System (response)

Shown above: Filtering for IP address 172.16.3.188 and finding it's an Apple product.

ip.addr eq 172.16.3.188 and http.request

Time	Src	port	Dst	port	Host	Info
2018-12-18 19:34...	172.16.3.188	53504	104.96.221.57	80	init-p01st.push.apple.com	GET /
2018-12-18 19:36...	172.16.3.188	53509	151.101.64.70	80	nymag.com	GET /
2018-12-18 19:36...	172.16.3.188	53519	172.217.2.98	80	www.googletagmanager.com	GET /
2018-12-18 19:36...	172.16.3.188	53520	172.217.2.98	80	pagead2.googlesyndication.com	GET /
2018-12-18 19:36...	172.16.3.188	53522	151.101.32.70	80	fonts.nymag.com	GET /
2018-12-18 19:36...	172.16.3.188	53523	23.74.28.41	80	b.scorecardresearch.com	GET /
2018-12-18 19:36...	172.16.3.188	53523	23.74.28.41	80	b.scorecardresearch.com	GET /
2018-12-18 19:36...	172.16.3.188	53524	31.13.80.12	80	connect.facebook.net	GET /
2018-12-18 19:47...	172.16.3.188	53528	104.96.221.80	80	init-p01st.push.apple.com	GET /
2018-12-18 19:50...	172.16.3.188	53529	52.85.142.7	80	static.chartbeat.com	GET /
2018-12-18 19:50...	172.16.3.188	53531	172.217.0.98	80	www.googleadservices.com	GET /
2018-12-18 19:50...	172.16.3.188	53532	18.232.243.41	80	ping.chartbeat.net	GET /

Shown above: Filtering on HTTP requests from 172.16.3.188.

Wireshark · Follow TCP Stream (tcp.stream eq 987) · 2018-12-18-traffic-analysis-exercise

GET / HTTP/1.1
Host: nymag.com
Upgrade-Insecure-Requests: 1
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
User-Agent: Mozilla/5.0 (iPhone; CPU iPhone OS 12_1_1 like Mac OS X)
AppleWebKit/605.1.15 (KHTML, like Gecko) version/12.0 Mobile/15E148 Safari/604.1
Accept-Language: en-us
Accept-Encoding: gzip, deflate
Connection: keep-alive

HTTP/1.1 200 OK
Server: Apache/2.2.15 (CentOS)

3 client pkts, 327 server pkts, 3 turns.

Shown above: Following a TCP stream for the HTTP GET request to nymag.com and finding this is an iPhone.

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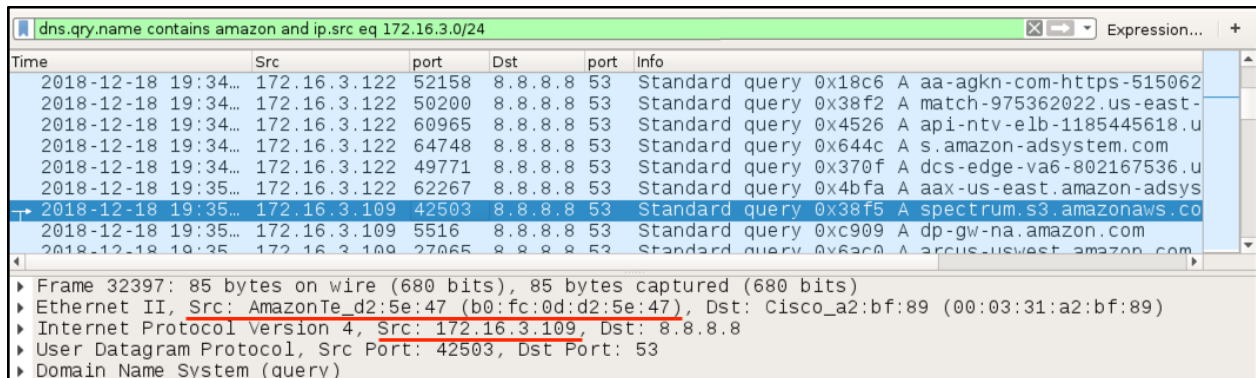
Q: Which IP address is mostly likely an Amazon Fire tablet?

Details: This one is a little more tricky than finding the previous answers. You'll find no HTTP traffic from an Amazon device in this pcap (it's all HTTPS). So how can we determine this answer? First, I recommend filtering on DNS requests or HTTPS domains with **amazon** in them. Amazon devices generate a great deal of traffic to Amazon domains. Use either of these two queries:

- **dns.qry.name contains amazon and ip.src eq 172.16.3.0/24**
- **ssl.handshake.type == 1 and ip contains amazon**

The search for DNS queries with "amazon" in the requested domain name reveals three IP addresses in the 172.16.3.0/24 block. The search for HTTPS domains with amazon reveals two IP addresses. Then check the MAC addresses for the associated 172.16.3.0/24 IP addresses from your results. The only one that has an Amazon MAC address is 172.16.3.109. That's the only Amazon device from the 172.16.3.0/24 block.

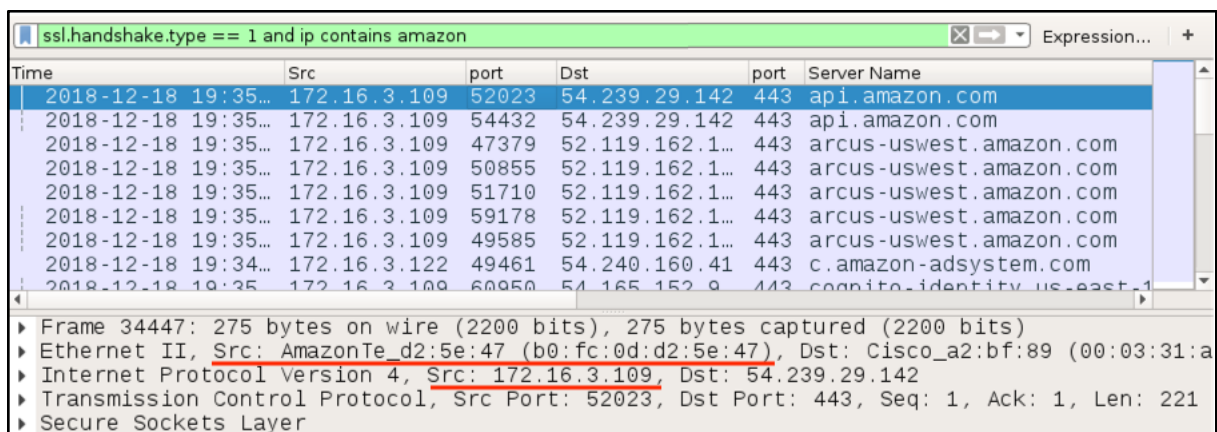
Here's another hint indicating this might be an Amazon Fire tablet (or at least some sort of Amazon device): The web browser on an Amazon Fire tablet is named **Silk**. You'll see a DNS request and HTTPS traffic to domains ending in **amazonsilk.com**.



Time	Src	port	Dst	port	Info
2018-12-18 19:34...	172.16.3.122	52158	8.8.8.8	53	Standard query 0x18c6 A aa-agkn-com-https-515062
2018-12-18 19:34...	172.16.3.122	50200	8.8.8.8	53	Standard query 0x38f2 A match-975362022.us-east-
2018-12-18 19:34...	172.16.3.122	60965	8.8.8.8	53	Standard query 0x4526 A api-ntv-elb-1185445618.u
2018-12-18 19:34...	172.16.3.122	64748	8.8.8.8	53	Standard query 0x644c A s.amazon-adsystem.com
2018-12-18 19:34...	172.16.3.122	49771	8.8.8.8	53	Standard query 0x370f A dcs-edge-va6-802167536.u
2018-12-18 19:35...	172.16.3.122	62267	8.8.8.8	53	Standard query 0x4bfa A aax-us-east.amazon-adsys
2018-12-18 19:35...	172.16.3.109	42503	8.8.8.8	53	Standard query 0x38f5 A spectrum.s3.amazonaws.co
2018-12-18 19:35...	172.16.3.109	5516	8.8.8.8	53	Standard query 0xc909 A dp-gw-na.amazon.com
2018-12-18 19:35...	172.16.3.109	27065	8.8.8.8	53	Standard query 0x6ac0 A arcus-uswest.amazon.com

Frame 32397: 85 bytes on wire (680 bits), 85 bytes captured (680 bits)
Ethernet II, Src: AmazonTe_d2:5e:47 (b0:fc:0d:d2:5e:47), Dst: Cisco_a2:bf:89 (00:03:31:a2:bf:89)
Internet Protocol Version 4, Src: 172.16.3.109, Dst: 8.8.8.8
User Datagram Protocol, Src Port: 42503, Dst Port: 53
Domain Name System (query)

Shown above: Filtering on DNS queries with "amazon" in the domain name and finding the Amazon device based on the MAC address.



Time	Src	port	Dst	port	Server Name
2018-12-18 19:35...	172.16.3.109	52023	54.239.29.142	443	api.amazon.com
2018-12-18 19:35...	172.16.3.109	54432	54.239.29.142	443	api.amazon.com
2018-12-18 19:35...	172.16.3.109	47379	52.119.162.1...	443	arcus-uswest.amazon.com
2018-12-18 19:35...	172.16.3.109	50855	52.119.162.1...	443	arcus-uswest.amazon.com
2018-12-18 19:35...	172.16.3.109	51710	52.119.162.1...	443	arcus-uswest.amazon.com
2018-12-18 19:35...	172.16.3.109	59178	52.119.162.1...	443	arcus-uswest.amazon.com
2018-12-18 19:35...	172.16.3.109	49585	52.119.162.1...	443	arcus-uswest.amazon.com
2018-12-18 19:34...	172.16.3.122	49461	54.240.160.41	443	c.amazon-adsystem.com
2018-12-18 19:35...	172.16.3.109	60950	54.165.152.9	443	cognito-identity-us-east-1

Frame 34447: 275 bytes on wire (2200 bits), 275 bytes captured (2200 bits)
Ethernet II, Src: AmazonTe_d2:5e:47 (b0:fc:0d:d2:5e:47), Dst: Cisco_a2:bf:89 (00:03:31:a2:bf:89)
Internet Protocol Version 4, Src: 172.16.3.109, Dst: 54.239.29.142
Transmission Control Protocol, Src Port: 52023, Dst Port: 443, Seq: 1, Ack: 1, Len: 221
Secure Sockets Layer

Shown above: Filtering on HTTPS traffic with "amazon" in the domain name and finding the Amazon device based on the MAC address.

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Time	Src	port	Dst	port	Server Name
2018-12-18 19:35...	172.16.3.109	33086	54.230.19.177	443	geo-ip-prod.service.amazonsilk
2018-12-18 19:35...	172.16.3.109	33376	54.86.6.8	443	r13s.service.amazonsilk.com
2018-12-18 19:35...	172.16.3.109	33377	54.86.6.8	443	r13s.service.amazonsilk.com
2018-12-18 19:35...	172.16.3.109	33378	54.86.6.8	443	r13s.service.amazonsilk.com
2018-12-18 19:35...	172.16.3.109	33379	54.86.6.8	443	r13s.service.amazonsilk.com

Frame 35316: 291 bytes on wire (2328 bits), 291 bytes captured (2328 bits)
Ethernet II, Src: AmazonTe_d2:5e:47 (b0:fc:0d:d2:5e:47), Dst: Cisco_a2:bf:89 (00:03:31:a...)
Internet Protocol Version 4, Src: 172.16.3.109, Dst: 54.230.19.177
Transmission Control Protocol, Src Port: 33086, Dst Port: 443, Seq: 1, Ack: 1, Len: 225
Secure Sockets Layer

Shown above: Filtering on HTTPS traffic with "silk" in the domain name.

Q: Which three IP addresses represent Windows hosts that connect to the domain controller at 172.16.3.2?

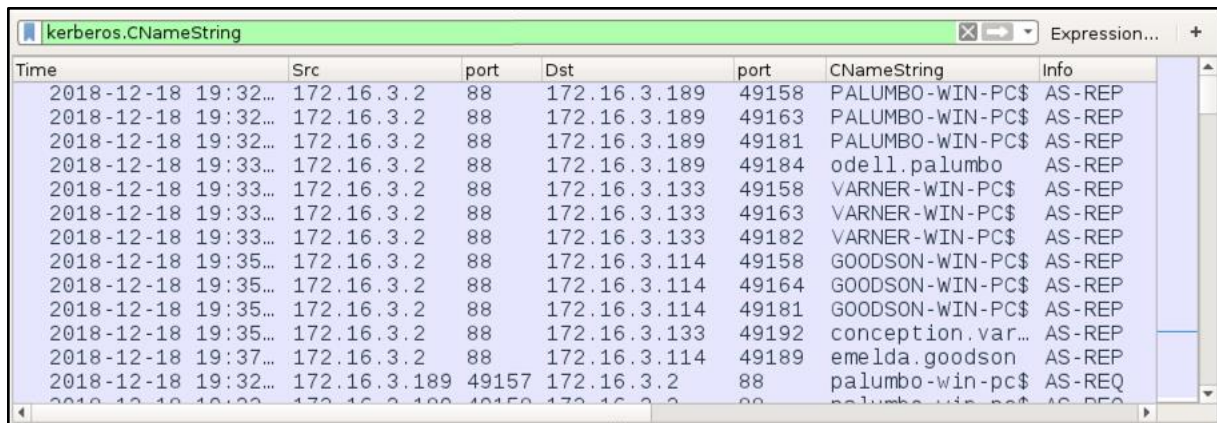
Details: This one's relatively easy. You can filter on **Kerberos.CNameString** and find the Windows hosts that logged into the Domain Controller. If you apply the CNameString value as a column, you can quickly find the host names and the user account names for these Windows hosts.

Time	Src	port	Dst
2018-12-18 19:32...	172.16.3.2	88	172.16.3.2
2018-12-18 19:32...	172.16.3.2	88	172.16.3.2
2018-12-18 19:32...	172.16.3.2	88	172.16.3.2
2018-12-18 19:33...	172.16.3.2	88	172.16.3.2
2018-12-18 19:33...	172.16.3.2	88	172.16.3.2
2018-12-18 19:33...	172.16.3.2	88	172.16.3.2

Frame 35206: 70 bytes on wire (560 bits)
Ethernet II, Src: Dell_39:42:bc (00:1a:a...)
Internet Protocol Version 4, Src: 172.16...
Transmission Control Protocol, Src Port: ...
[2 Reassembled TCP Segments (1476 bytes)]
Kerberos
Record Mark: 1472 bytes
tgs-rep
pvno: 5
msg-type: krb-tgs-rep (13)
crealm: EGGNOGSOUP.COM
cname
name-type: KRB5-NT-PRINCIPAL (1)
cname-string: 1 item
CNameString: GOODSON-WIN-PC\$
ticket
enc-part

Shown above: Filtering on Kerberos.CNameString and applying the CNameString value as a column.

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The image shows a Wireshark packet capture of Kerberos traffic. The filter is 'kerberos.CNameString'. The table below represents the data shown in the packet list pane.

Time	Src	port	Dst	port	CNameString	Info
2018-12-18 19:32...	172.16.3.2	88	172.16.3.189	49158	PALUMBO-WIN-PC\$	AS-REP
2018-12-18 19:32...	172.16.3.2	88	172.16.3.189	49163	PALUMBO-WIN-PC\$	AS-REP
2018-12-18 19:32...	172.16.3.2	88	172.16.3.189	49181	PALUMBO-WIN-PC\$	AS-REP
2018-12-18 19:33...	172.16.3.2	88	172.16.3.189	49184	odell.palumbo	AS-REP
2018-12-18 19:33...	172.16.3.2	88	172.16.3.133	49158	VARNER-WIN-PC\$	AS-REP
2018-12-18 19:33...	172.16.3.2	88	172.16.3.133	49163	VARNER-WIN-PC\$	AS-REP
2018-12-18 19:33...	172.16.3.2	88	172.16.3.133	49182	VARNER-WIN-PC\$	AS-REP
2018-12-18 19:35...	172.16.3.2	88	172.16.3.114	49158	GOODSON-WIN-PC\$	AS-REP
2018-12-18 19:35...	172.16.3.2	88	172.16.3.114	49164	GOODSON-WIN-PC\$	AS-REP
2018-12-18 19:35...	172.16.3.2	88	172.16.3.114	49181	GOODSON-WIN-PC\$	AS-REP
2018-12-18 19:35...	172.16.3.2	88	172.16.3.133	49192	conception.var...	AS-REP
2018-12-18 19:37...	172.16.3.2	88	172.16.3.114	49189	emelda.goodson	AS-REP
2018-12-18 19:32...	172.16.3.189	49157	172.16.3.2	88	palumbo-win-pc\$	AS-REQ

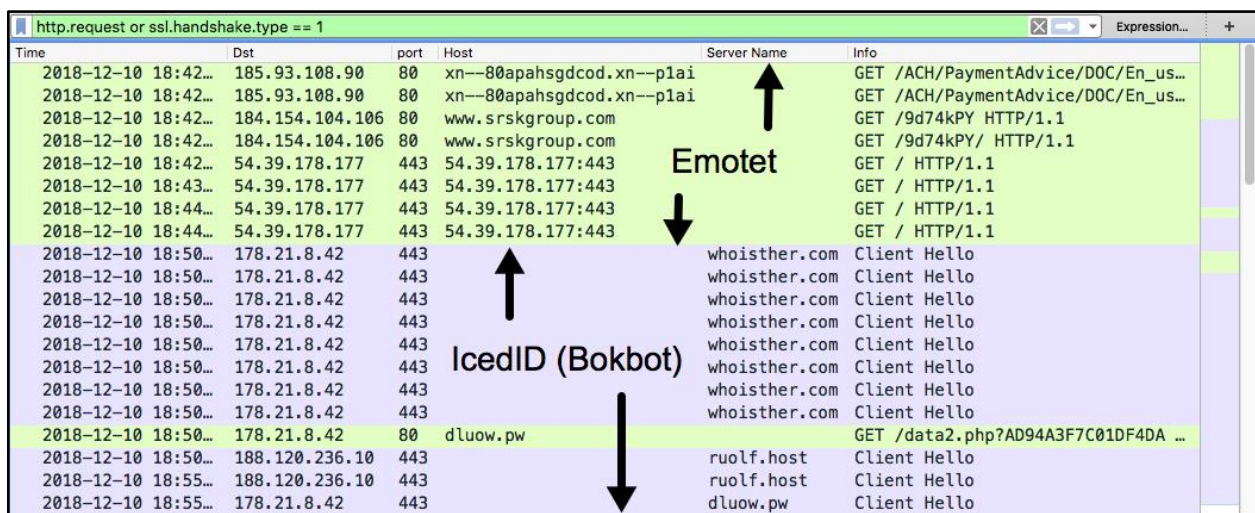
Shown above: With CNameString as a column, you can quickly find the Windows hosts and user accounts that logged in to the Domain Controller.

Q: Which of the three Windows hosts shows indications of an infection with Emotet and IcedID banking Trojan (Bokbot)?

Details: To answer this question, first we should review a recent example of Emotet and IcedID traffic. The most recent one I published is here:

- <https://www.malware-traffic-analysis.net/2018/12/10/index2.html>

The image from that blog post shows what the traffic looks like:



The image shows a Wireshark packet capture of HTTP traffic. The filter is 'http.request or ssl.handshake.type == 1'. The table below represents the data shown in the packet list pane. Annotations include arrows pointing to specific traffic flows: 'Emotet' points to traffic from 54.39.178.177 to 54.39.178.177:443, and 'IcedID (Bokbot)' points to traffic from 178.21.8.42 to 178.21.8.42:443.

Time	Dst	port	Host	Server Name	Info
2018-12-10 18:42...	185.93.108.90	80	xn--80apahsgdcod.xn--p1ai		GET /ACH/PaymentAdvice/DOC/En_us...
2018-12-10 18:42...	185.93.108.90	80	xn--80apahsgdcod.xn--p1ai		GET /ACH/PaymentAdvice/DOC/En_us...
2018-12-10 18:42...	184.154.104.106	80	www.srsgroup.com		GET /9d74kPY HTTP/1.1
2018-12-10 18:42...	184.154.104.106	80	www.srsgroup.com		GET /9d74kPY HTTP/1.1
2018-12-10 18:42...	54.39.178.177	443	54.39.178.177:443		GET / HTTP/1.1
2018-12-10 18:43...	54.39.178.177	443	54.39.178.177:443		GET / HTTP/1.1
2018-12-10 18:44...	54.39.178.177	443	54.39.178.177:443		GET / HTTP/1.1
2018-12-10 18:44...	54.39.178.177	443	54.39.178.177:443		GET / HTTP/1.1
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	443		whoisther.com	Client Hello
2018-12-10 18:50...	178.21.8.42	80	dluow.pw		GET /data2.php?AD94A3F7C01DF4DA ...
2018-12-10 18:50...	188.120.236.10	443		ruolf.host	Client Hello
2018-12-10 18:55...	188.120.236.10	443		ruolf.host	Client Hello
2018-12-10 18:55...	178.21.8.42	443		dluow.pw	Client Hello

Shown above: Image from the Emotet and IcedID infection traffic on 2018-12-10.

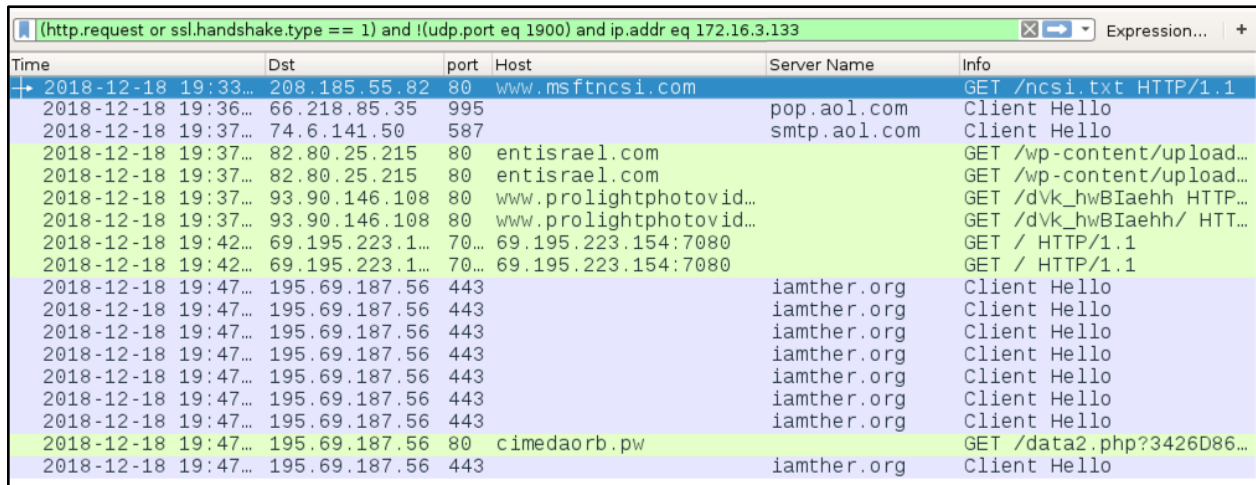
We have the three Windows IP addresses, so we should filter on web traffic and each IP address:

- (http.request or ssl.handshake.type == 1) and !(ssdp) and ip.addr eq 172.16.3.114
- (http.request or ssl.handshake.type == 1) and !(ssdp) and ip.addr eq 172.16.3.133
- (http.request or ssl.handshake.type == 1) and !(ssdp) and ip.addr eq 172.16.3.189

Note: I generally use !(udp.port 1900) instead of !(ssdp), but the results in this case should be the same.

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Note how the IP address with traffic that looks closest to the previous image (follows the same general patterns) is 172.16.3.133.



Time	Dst	port	Host	Server Name	Info
2018-12-18 19:33...	208.185.55.82	80	www.msftncsi.com		GET /ncsi.txt HTTP/1.1
2018-12-18 19:36...	66.218.85.35	995		pop.aol.com	Client Hello
2018-12-18 19:37...	74.6.141.50	587		smtp.aol.com	Client Hello
2018-12-18 19:37...	82.80.25.215	80	entisrael.com		GET /wp-content/upload...
2018-12-18 19:37...	82.80.25.215	80	entisrael.com		GET /wp-content/upload...
2018-12-18 19:37...	93.90.146.108	80	www.prolightphotovid...		GET /dvk_hwBIaehh HTTP...
2018-12-18 19:37...	93.90.146.108	80	www.prolightphotovid...		GET /dvk_hwBIaehh/ HTT...
2018-12-18 19:42...	69.195.223.1...	70...	69.195.223.154:7080		GET / HTTP/1.1
2018-12-18 19:42...	69.195.223.1...	70...	69.195.223.154:7080		GET / HTTP/1.1
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello
2018-12-18 19:47...	195.69.187.56	80	cimedaorb.pw		GET /data2.php?3426D86...
2018-12-18 19:47...	195.69.187.56	443		iamther.org	Client Hello

Shown above: Web traffic filtered on 172.16.3.133.

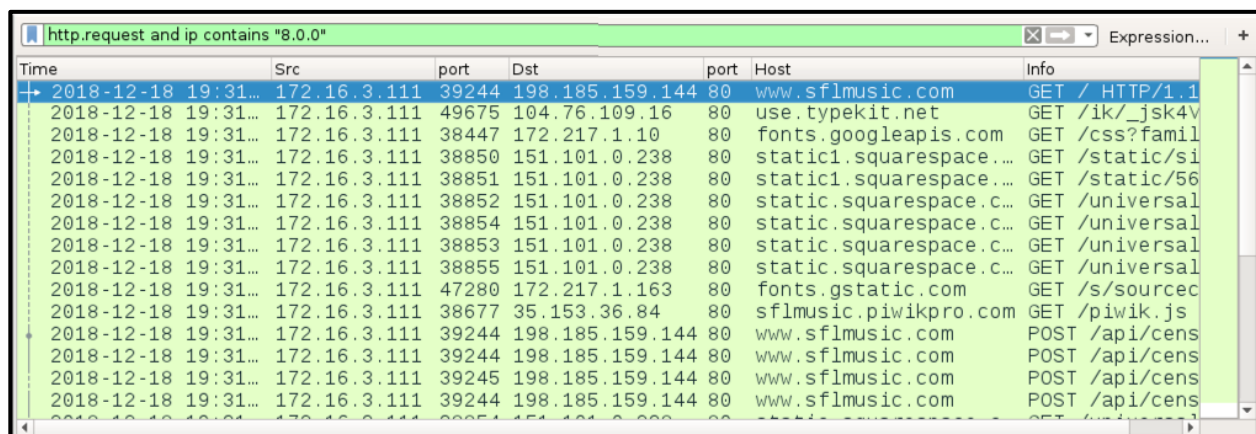
Q: Which IP address is a host running Android 8.0.0?

Q: What is the brand and model of the phone running Android 8.0.0?

Details: We can quickly answer this if there is regular HTTP traffic with a User-Agent string that has "8.0.0" in it. Fortunately, we're in luck, because there is! Use the following search, making sure to put 8.0.0 in quotation marks:

- **http.request and ip contains "8.0.0"**

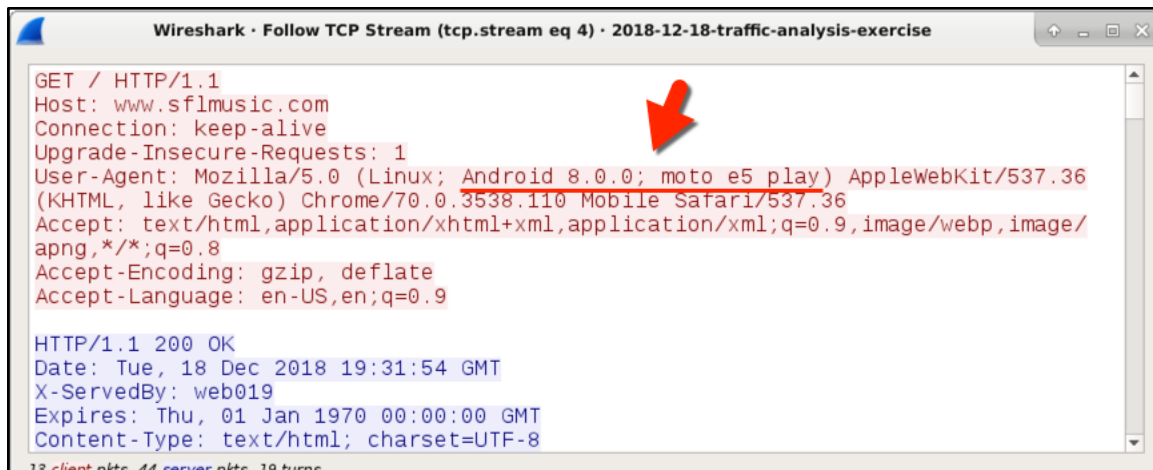
That should reveal 172.16.3.111 as the only internal IP address in the results. Follow the TCP stream for the first HTTP GET request, and you can confirm it's running Android 8.0.0 according to the User-Agent string in the HTTP request headers. You'll also find the device name in the HTTP request headers: **moto e5 play**



Time	Src	port	Dst	port	Host	Info
2018-12-18 19:31...	172.16.3.111	39244	198.185.159.144	80	www.sflmusic.com	GET / HTTP/1.1
2018-12-18 19:31...	172.16.3.111	49675	104.76.109.16	80	use.typekit.net	GET /ik/_jsk4v
2018-12-18 19:31...	172.16.3.111	38447	172.217.1.10	80	fonts.googleapis.com	GET /css?family=
2018-12-18 19:31...	172.16.3.111	38850	151.101.0.238	80	static1.squarespace...	GET /static/si
2018-12-18 19:31...	172.16.3.111	38851	151.101.0.238	80	static1.squarespace...	GET /static/56
2018-12-18 19:31...	172.16.3.111	38852	151.101.0.238	80	static.squarespace.c...	GET /universal
2018-12-18 19:31...	172.16.3.111	38854	151.101.0.238	80	static.squarespace.c...	GET /universal
2018-12-18 19:31...	172.16.3.111	38853	151.101.0.238	80	static.squarespace.c...	GET /universal
2018-12-18 19:31...	172.16.3.111	38855	151.101.0.238	80	static.squarespace.c...	GET /universal
2018-12-18 19:31...	172.16.3.111	47280	172.217.1.163	80	fonts.gstatic.com	GET /s/sourcec
2018-12-18 19:31...	172.16.3.111	38677	35.153.36.84	80	sflmusic.piwikpro.com	GET /piwik.js
2018-12-18 19:31...	172.16.3.111	39244	198.185.159.144	80	www.sflmusic.com	POST /api/cens
2018-12-18 19:31...	172.16.3.111	39244	198.185.159.144	80	www.sflmusic.com	POST /api/cens
2018-12-18 19:31...	172.16.3.111	39245	198.185.159.144	80	www.sflmusic.com	POST /api/cens
2018-12-18 19:31...	172.16.3.111	39244	198.185.159.144	80	www.sflmusic.com	POST /api/cens

Shown above: One way to search for a device running Android 8.0.0.

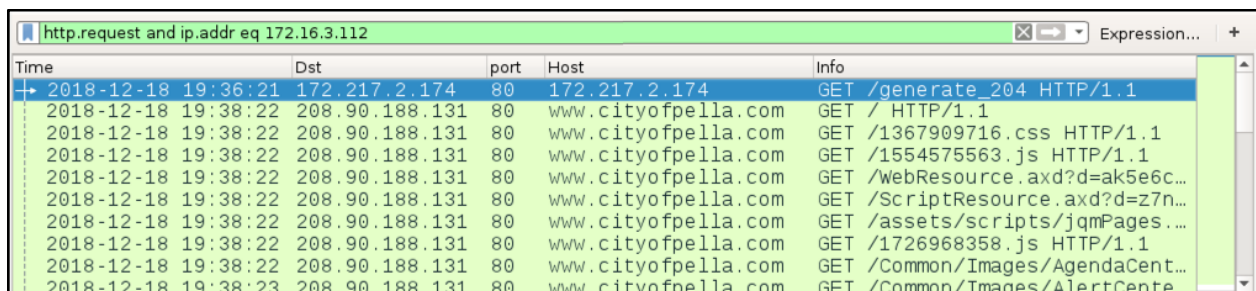
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Shown above: Confirming the device is running Android 8.0.0 and finding out it is a Moto E5 Play.

Q: What is the brand and type of device on 172.16.3.112?

Details: This one is another where we're fortunate to have regular HTTP traffic so we can find a User-Agent line in the HTTP request headers. In this case, the device is running Android 4.4.4 and has a model number of SM-T113. A quick Google search on SM-T113 reveals this device is a Samsung Galaxy Tab E Lite.

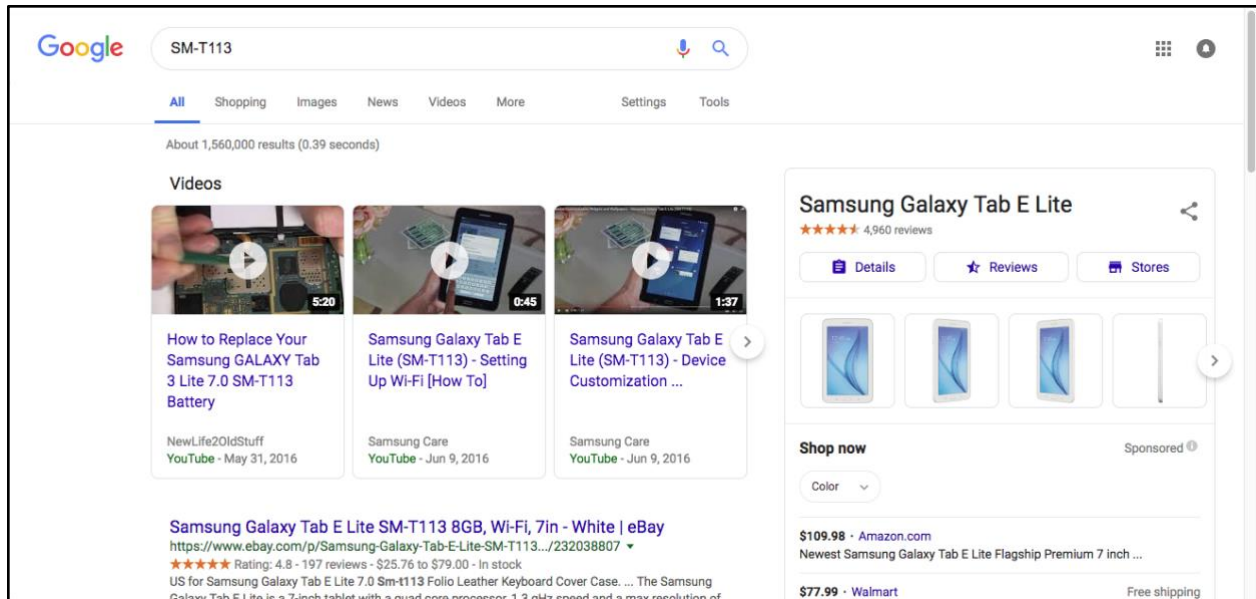


Shown above: Filtering on HTTP requests from 172.16.3.112.



Shown above: Seeing SM-T113 as the model number from the HTTP request headers.

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Shown above: Google search on SM-T113 reveals it's a Samsung Galaxy Tab E Lite.

FINAL WORDS

That answers all of the questions in this month's exercise. Bear in mind these answers are how I got to the information, and there is usually more than one method to investigate these questions.

Until next month!